

Organizations Online Fitness Coaching Platform

Chapter 1. Business Modelling

1a.

1.Business actors:

Customers (people that are interested in fitness), fitness coaches

2.Business processes:

1.User registration

Inputs:

- User email and password(for account creation)
- Users fitness goals
- Users info (experience and schedule)

Methods:

- User enters registration information
- System validates the data
- Account gets made with the info given

Outputs:

- Registered account

2. Coach Matching

Inputs

- User fitness profile
- Coach database
- Coach specializations

Method

- System analyzes user fitness goals
- System searches available coaches
- System filters coaches by expertise
- User selects a coach

Outputs

- User-coach assignment
- Coaching relationship created

3. Workout Plan Creation and Delivery

Inputs

- User fitness profile
- User goals
- Coach expertise

Method

- Coach reviews user profile
- Coach designs workout plan
- Coach uploads exercises and instructions
- System publishes the plan to the user

Outputs

- Personalized workout plan
- Training schedule available to the user

4. Progress Tracking

Inputs

- Workout logs
- Performance metrics

Method

- User records workout sessions
- System collects performance data
- System analyzes user progress
- System generates reports

Outputs

- Progress reports

- Fitness statistics

3.Business agents:

Platform administrator (upkeep)

Fitness coaches

Nutrition specialists

Data analyst

Advertising team

4.Business objects:

Workout Log

Nutrition Plan

User Profile

Progress Report

Community Post

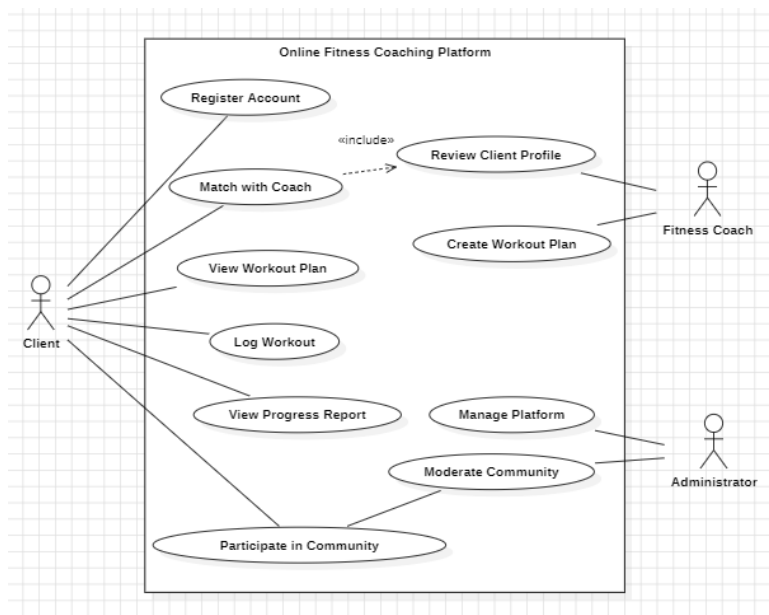
Workout Plan

Coach Profile

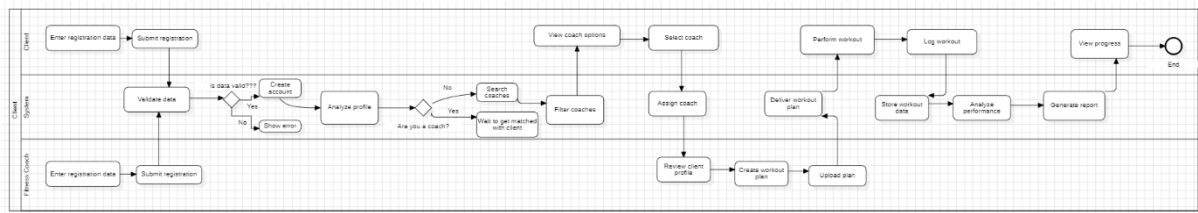
5. Business rules:

- 1) A user must register an account before accessing coaching services
- 2) Only certified coaches can create workout plans
- 3) Each user may be assigned to one primary coach at a time
- 4) Workout plans must include exercise instructions and schedule
- 5) Users must log workout sessions to update progress metrics
- 6) Nutrition recommendations must be aligned with user fitness goals
- 7) Community posts must follow platform moderation rules

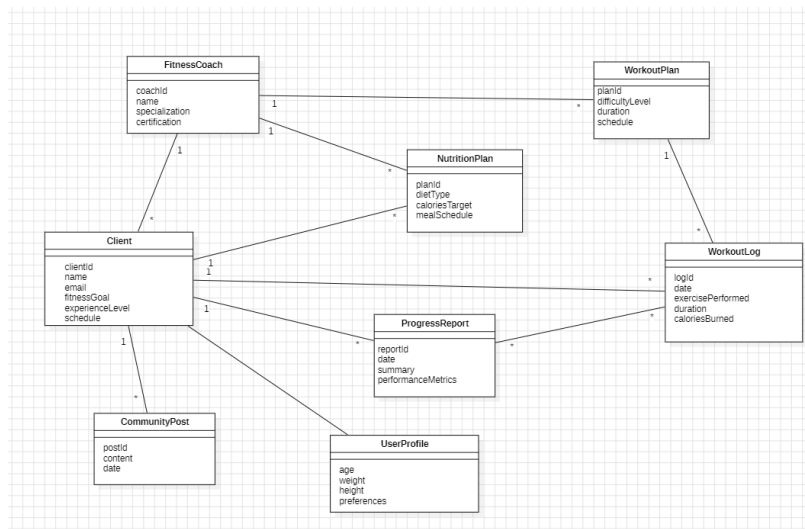
1b. Business Use case diagram



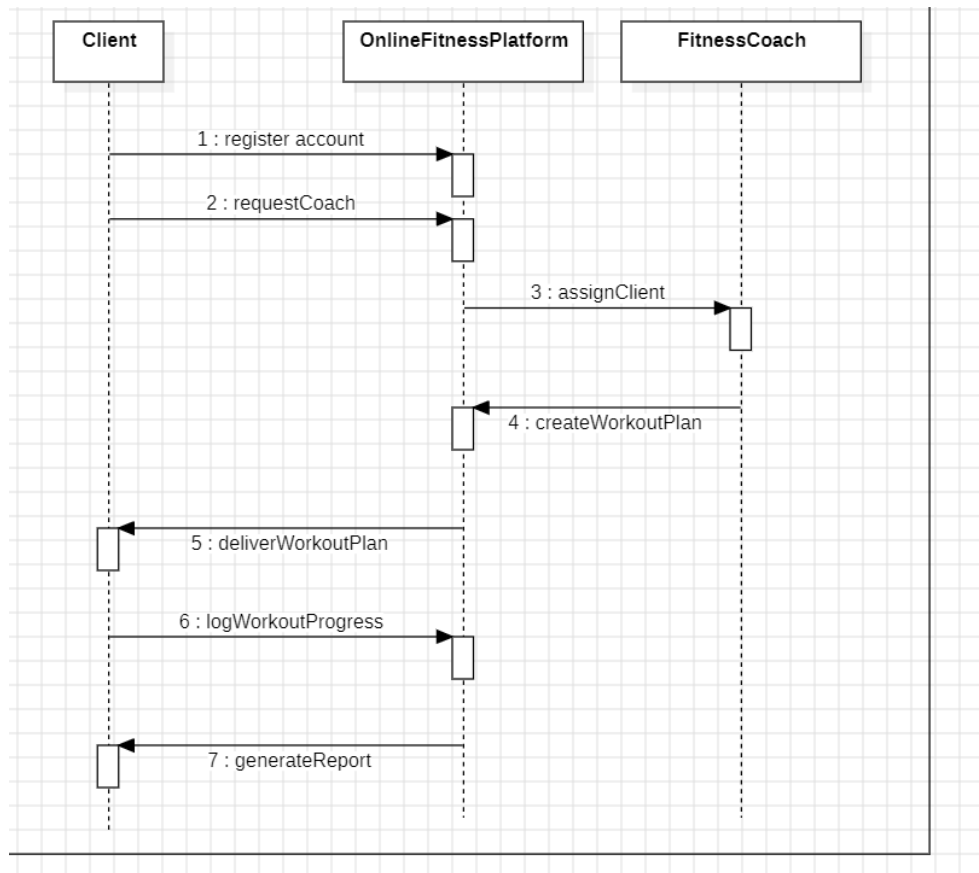
BPNM



1d. Business domain model

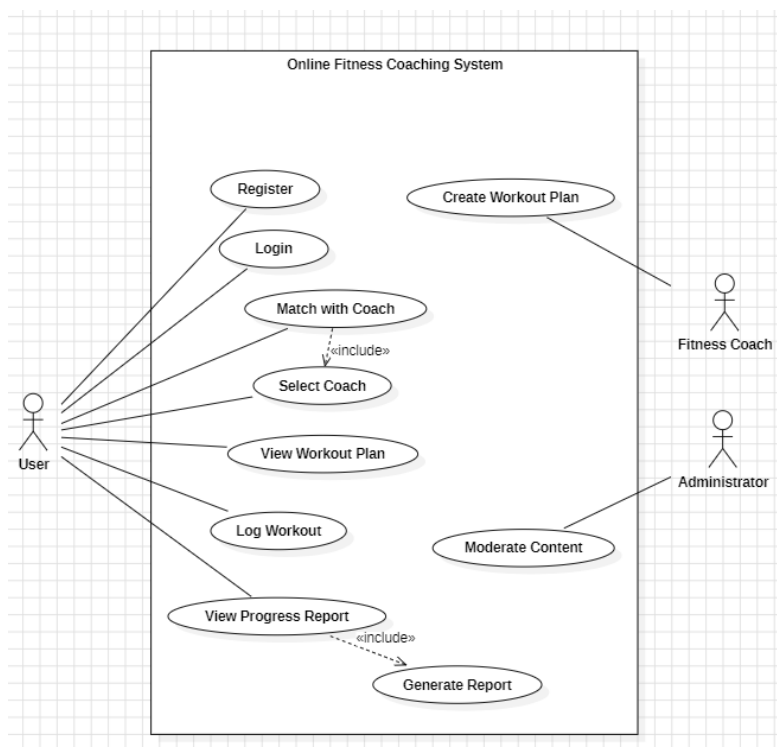


Business Sequence Diagram



Chapter 2. User Requirements

2a. Software use case diagram



2b. Use cases

The system has to register users.

The system has to authenticate users.

The system has to match users with fitness coaches.

The system has to assign a coach to a user.

The system has to provide workout plans to users.

The system has to store workout logs.

The system has to generate progress reports.

The system has to allow coaches to create workout plans.

The system has to allow administrators to moderate community content.

The system has to allow users to participate in community discussions.

2c. Non-Functional Requirements

NFR1. Usability: The system must provide an intuitive and user-friendly interface for both users and fitness coaches.

NFR2. Performance: The system must respond to user actions within a maximum of 2 seconds.

NFR3. Reliability: The system must ensure continuous operation with minimal downtime and no data loss.

NFR4. Security: The system must ensure secure authentication and protect user data through encryption.

NFR5. Scalability: The system must support a growing number of users and fitness coaches without performance degradation.

NFR6. Availability: The system must be accessible 24/7 for users and coaches.

NFR7. Maintainability: The system must allow easy updates, bug fixes, and feature enhancements.

NFR8. Portability: The system must be accessible via web browsers and mobile devices.

NFR9. Accuracy: The system must provide correct matching results and accurate progress reports.

NFR10. Robustness: The system must handle invalid inputs and unexpected errors w

2d. Use Case Descriptions

Use Case: Register User

Actor: User

Description:

The system allows a new user to create an account by providing personal and fitness-related information.

Preconditions:

- User is not registered

Main Flow:

1. User enters registration data (email, password, fitness info)
2. System validates the data
3. System creates the user account
4. System confirms successful registration

Alternative Flow:

A1: Invalid data

- System displays error and requests correction

A2: Email already exists

- System prompts user to log in

Postconditions:

- User account is successfully created

Non-functional Requirements:

- Usability: The registration process must be simple and intuitive
- Performance: System response time should be under 2 seconds
- Security: User data must be encrypted and securely stored

- Robustness: System must handle invalid inputs without failure
- Platform: Accessible via web and mobile devices

Use Case: Match with Coach

Actor: User

Description:

The system matches the user with suitable fitness coaches based on their profile and goals.

Preconditions:

- User is registered and logged in

Main Flow:

1. User requests coach matching
2. System analyzes user profile
3. System searches available coaches
4. System filters coaches by specialization
5. System displays matching coaches
6. User selects a coach
7. System assigns the coach

Alternative Flow:

A1: No available coaches

- System notifies user

A2: User cancels selection

- Process stops

Postconditions:

- User is assigned to a fitness coach

Non-functional Requirements:

- Performance: Matching results should be generated quickly
- Accuracy: Recommendations must be relevant to user goals
- Scalability: System must support many users and coaches
- Robustness: System must handle cases with no available coaches

Use Case: Create Workout Plan

Actor: Fitness Coach

Description:

The system allows the coach to create a personalized workout plan for a client.

Preconditions:

- Coach is authenticated
- Client is assigned to the coach

Main Flow:

1. Coach accesses client profile
2. Coach designs workout plan
3. Coach enters exercises and schedule
4. System saves the workout plan
5. System makes the plan available to the user

Alternative Flow:

A1: Invalid input

- System displays error

A2: Coach cancels creation

- Plan is not saved

Postconditions:

- Workout plan is successfully stored and available to the user

Non-functional Requirements:

- Usability: Interface must allow easy plan creation
- Performance: Plan saving should be fast
- Reliability: Data must not be lost
- Platform: Accessible via web platform

Use Case: Log Workout

Actor: User

Description:

The system allows the user to log completed workouts and track performance.

Preconditions:

- User has an assigned workout plan

Main Flow:

1. User selects workout session
2. User enters workout data (duration, exercises, etc.)
3. System validates input
4. System stores workout data
5. System updates user progress

Alternative Flow:

A1: Invalid input

- System displays error

A2: System failure

- User retries logging

Postconditions:

- Workout data is saved
- Progress is updated

Non-functional Requirements:

- Usability: Logging workouts must be quick and simple
- Performance: Data should be stored instantly
- Reliability: No data loss allowed
- Availability: System must be accessible at all times

Use Case: View Progress Report

Actor: User

Description:

The system provides users with reports on their fitness progress.

Preconditions:

- User has logged workout data

Main Flow:

1. User requests progress report
2. System retrieves workout data
3. System analyzes performance
4. System generates progress report
5. System displays report to user

Alternative Flow:

A1: No data available

- System notifies user

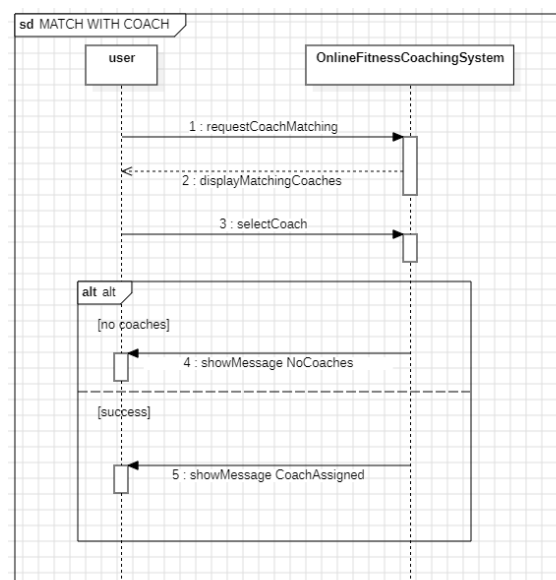
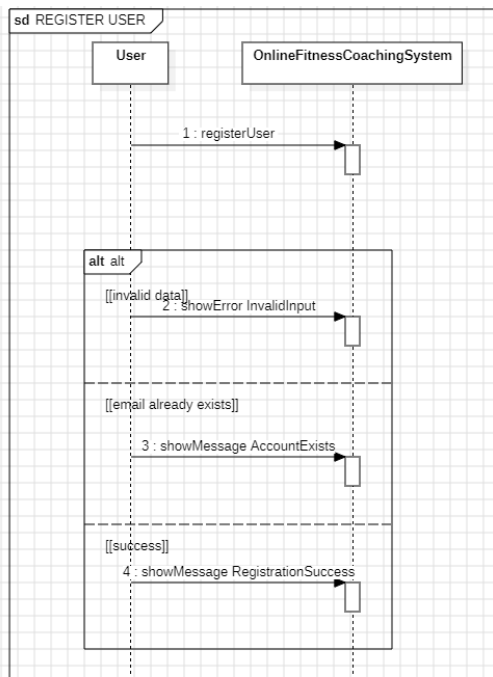
Postconditions:

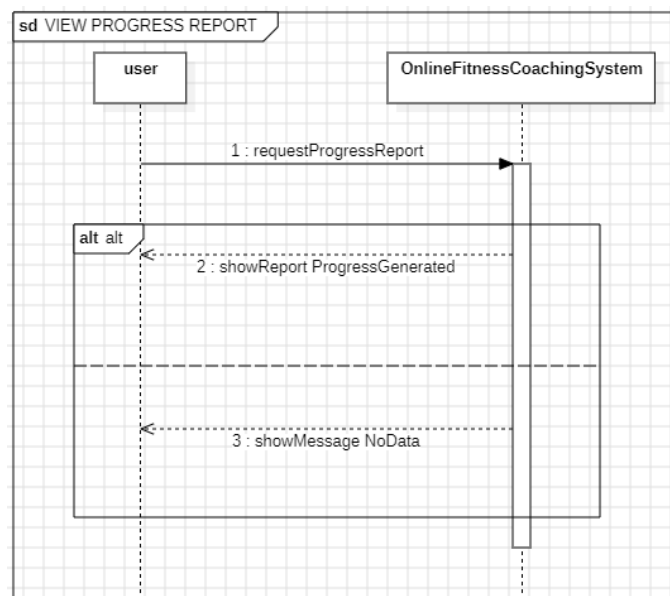
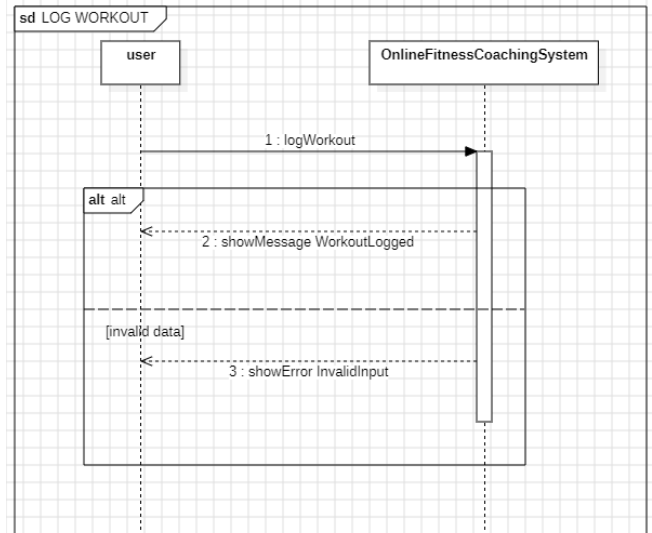
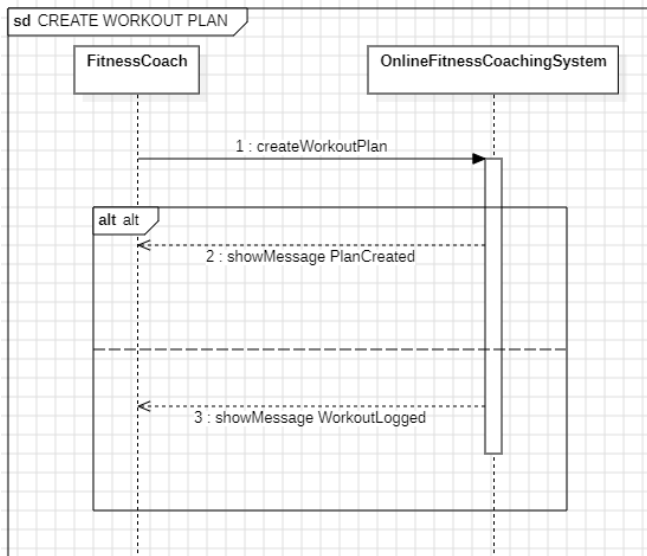
- User receives updated progress report

Non-functional Requirements:

- Performance: Reports must be generated quickly
- Accuracy: Data analysis must be correct
- Usability: Reports must be easy to understand
- Reliability: Data must be consistent and correct

2e. System Sequence Diagrams





2f. Operation contracts (for 2 system operations)

Operation Contract 1

Operation: registerUser(email, password, fitnessData)

Preconditions:

- The user is not already registered
- Required fields (email, password) are provided

Postconditions:

- A new user account is created
- User data is stored in the system
- The user is added to the user database

Operation Contract 2

Operation: assignCoach(userID, coachID)

Preconditions:

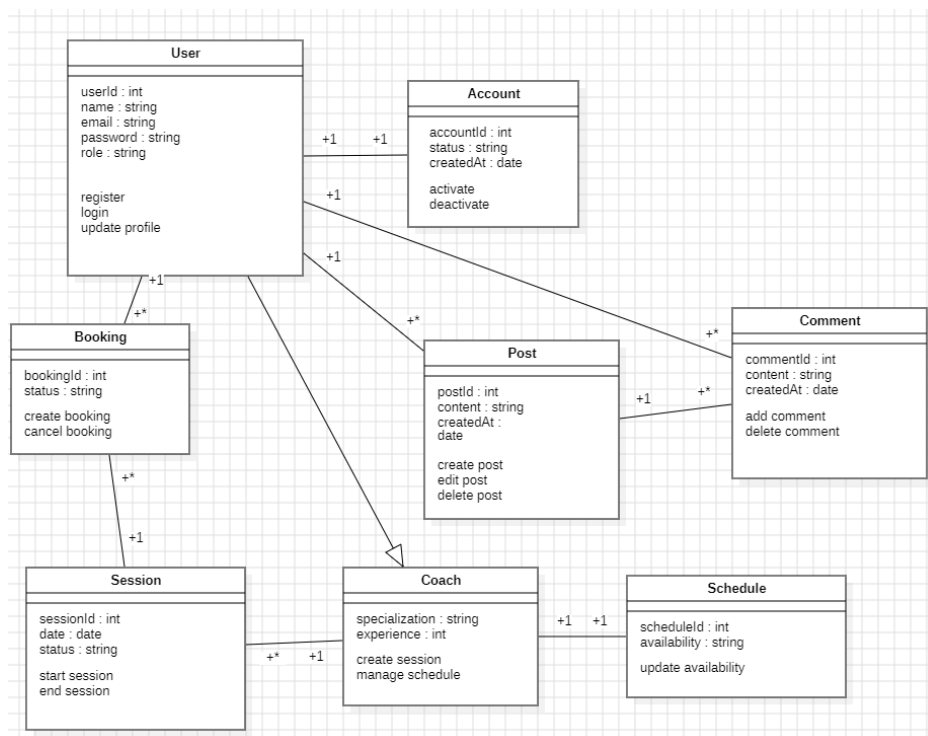
- The user is registered and logged in
- The selected coach is available

Postconditions:

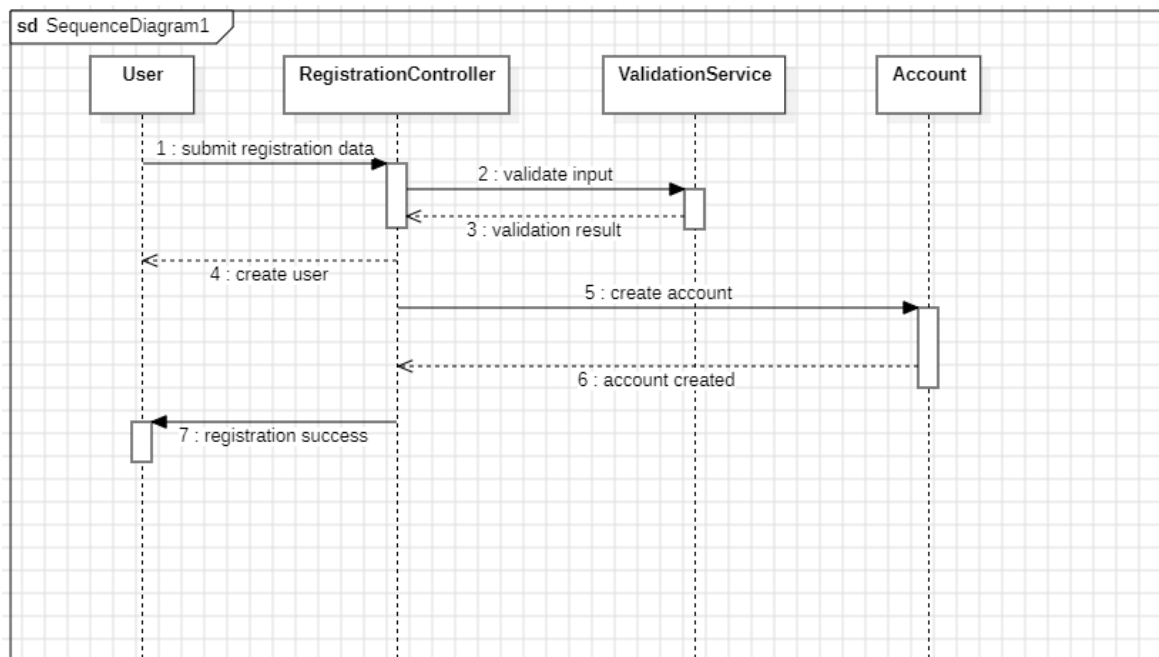
- The user is assigned to the selected coach
- A coaching relationship is created
- The assignment is stored in the system

Chapter 3. Design

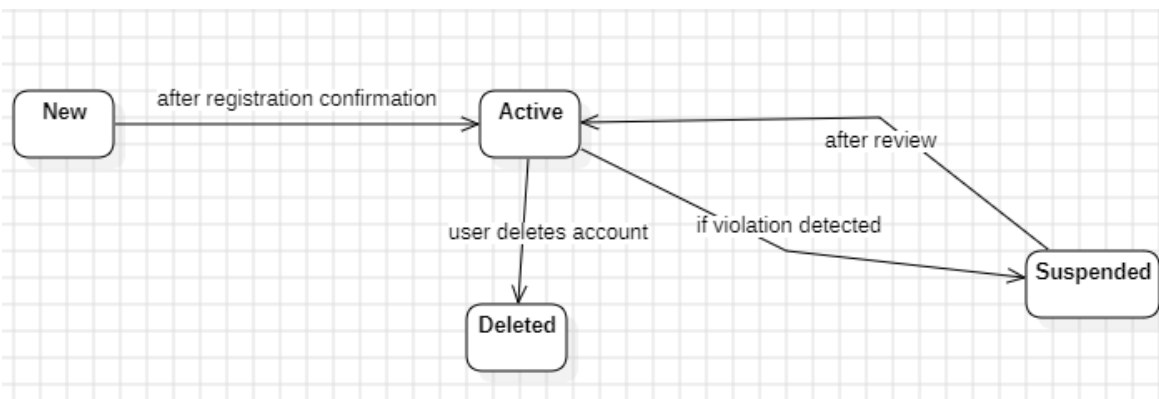
3a. Design class diagram (each software use case)



3b. Sequence diagram (each software use case)



3c. Statechart diagram (one object)



3d. Software architecture

Software Architecture

The system follows a 3-layer architecture:

1. Presentation Layer
 - User interface (web/mobile)
 - Handles user input and displays output
2. Application Layer (Business Logic)
 - Controllers (e.g., RegistrationController)

- Services (ValidationService, CoachingService)
- Implements use case logic

3. Data Layer

- Database (Users, Accounts, Sessions)
- Data access objects (DAO)

Architecture Flow:

User → UI → Controller → Service → Database

Benefits:

- Separation of concerns
- Easier maintenance
- Scalability